

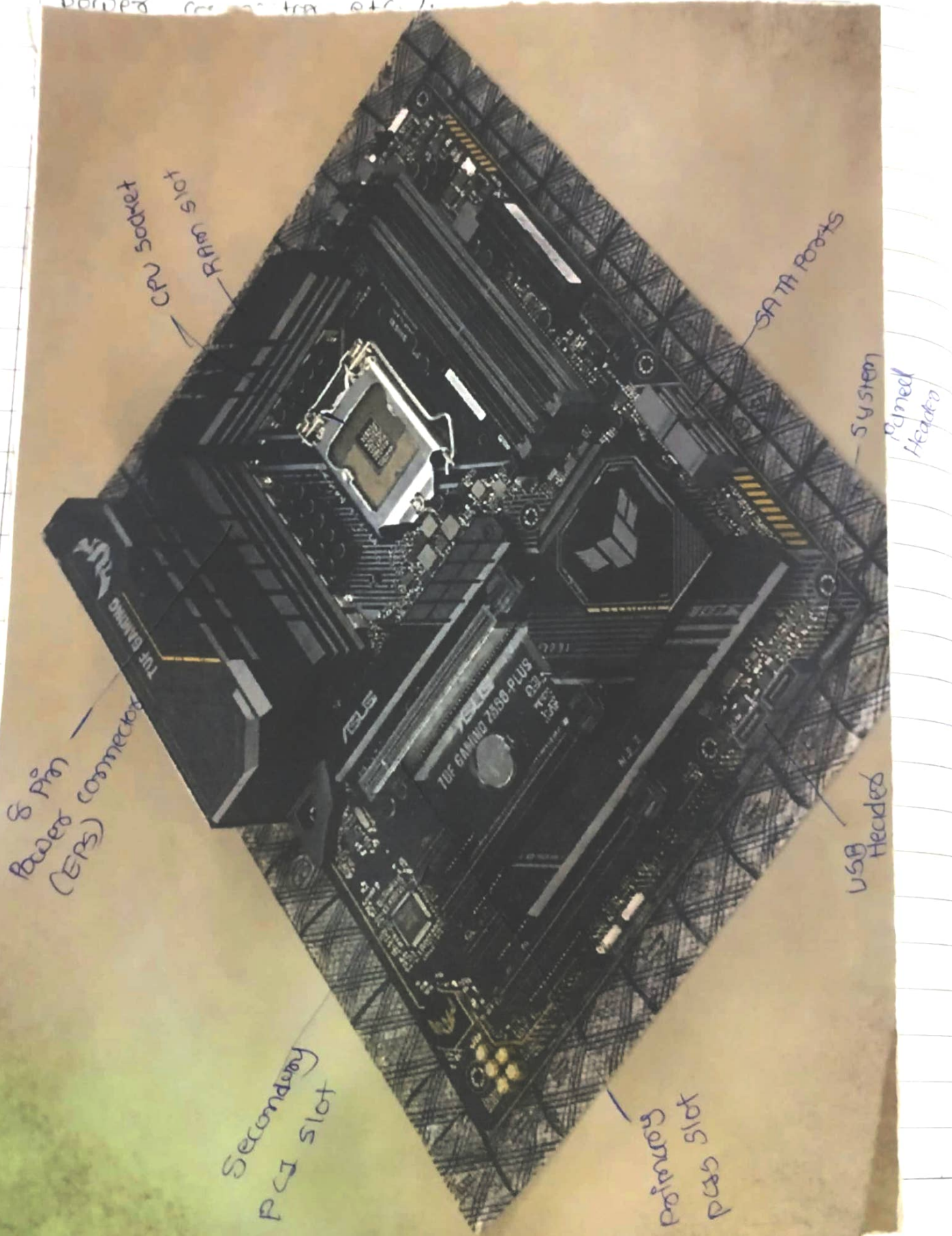
Assignment Day-2

Page No.

Date

Day 2:

1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM, PCIe slots, SATA ports, USB Headers, CPU Socket, power connector etc.).



2. Research and list 3 popular CPU socket types (For Example :- LGA 1151, AM4, LGA 1200) and name one compatible CPU model for each.

→

1. AM4 :-

AM4 is widely used socket platform designed for DDR4 memory system. It supported multiple generations of budget desktop processors and remains a popular choice for budget friendly PC builds.

compatible CPU :- AMD Ryzen 5 5600X / AMD Ryzen 7 5800X3D

2. LGA 1200 :-

It's introduced by Intel to connect desktop processors to the motherboard. The 1200 refers to the exact number of pins inside the socket. It's work with 10th generation & 11th generation Intel core processors. Systems using this socket rely on DDR4 RAM and support PCIe connections for graphics cards and fast storage.

compatible CPU :- Intel core i5 - 10400 (10th gen).
Intel core i7 - 11700K (11th gen).

3. LGA 1700 (Intel) :-

Intel's mainstream socket platform designed for its 12th, 13th and 14th generation Intel core processors. It's depending on the motherboard, it works with both DDR4 & DDR5 memory.

Compatible CPU :- Intel core i5-13600K (13th gen)
Intel core i7-14700K (14th gen)

Task 3. Explain in your words the primary functions of the CPU and the chipset on a motherboard and describe how they work together when you open a music app like Spotify.

→ * CPU (Central processing unit) :-

CPU is "brain" of computer. It calculates numbers, runs code, processes data and makes decisions for every program which we use.

* Chipset :-

It is "traffic controller" built into the motherboard. It manages the flow of data between the CPU, memory (RAM), storage drives, sound card and external device (For ex:- mouse, keyboard, USB, headphones)

- How they work when we open Spotify :-
1. Click on the icon. The chipset receives this signal through the USB port and sends a message to the CPU saying. "The user wants to open Spotify."
 2. Loading the App: The CPU tells the storage drive (SSD/HDD) to send Spotify Program files to RAM. The chipset routes this data from the storage drive over to the memory.
 3. The CPU reads Spotify's code from the RAM, builds the app's interface, connects to the internet to load your playlists and prepare the music file.
 4. When you hit play the CPU processes the audio file and hands the sound data to the chipset. The chipset directs those audio signals to your speakers, audio jack or Bluetooth adapter (headphones) so you can hear the audio (song).

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different parts or slots by circling them and labeling each one.

